The update step in a Gaussian filter

In the update step in Gaussian filters, instead of directly computing the moments of

$$\begin{aligned} p(\mathbf{x}_k|\mathbf{y}_{1:k}) &= p(\mathbf{x}_k|\mathbf{y}_k, \mathbf{y}_{1:k-1}) \\ &\propto p(\mathbf{y}_k|\mathbf{x}_k)\mathcal{N}(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}), \end{aligned}$$

we first approximate the predicted joint distribution,

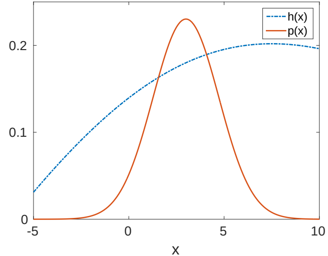
$$p(\mathbf{x}_k, \mathbf{y}_k|\mathbf{y}_{1:k-1})$$

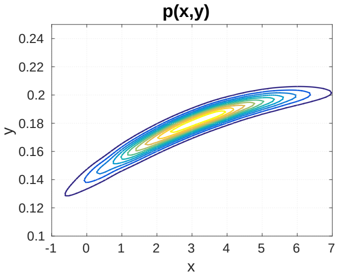
as a Gaussian and then compute the required conditional distribution, i.e., the filtering distribution.

For the non-linear measurement model

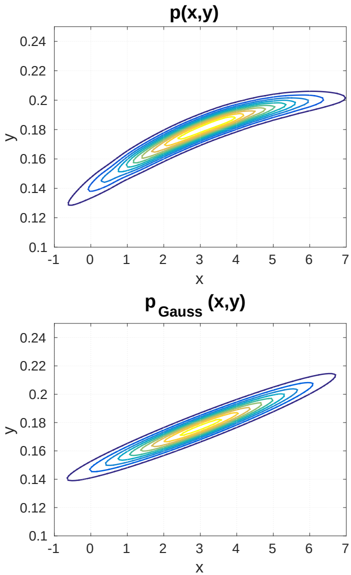
$$y = h(x) + r, \; r \sim \mathcal{N}(0, R)$$

shown to the right (top), the true joint density is depicted in the second figure (bottom).

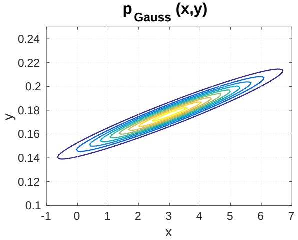




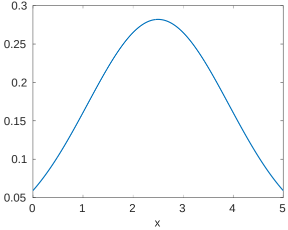
The true joint density $p(x, y)$, illustrated to the right (top), when approximated using moment matching, yields a Gaussian density (bottom).



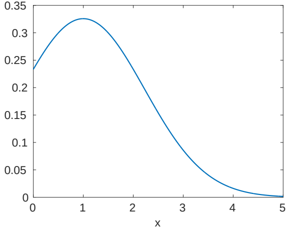
What is $p(x|y = 0.17)$?



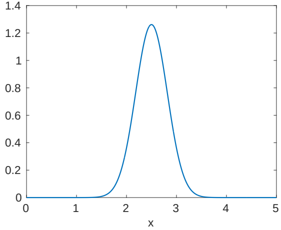
Yellow



Pink



Green



In the previous example, we have seen that the approximation to $p(\mathbf{x}_k|\mathbf{y}_{1:k})$ can be obtained from the Gaussian approximation to the joint density, $p(\mathbf{x}_k, \mathbf{y}_k|\mathbf{y}_{1:k-1})$. If we are given the predicted density $p(\mathbf{x}_k|\mathbf{y}_{1:k-1})$, which *other* parameters are required to describe the joint Gaussian density, $p(\mathbf{x}_k, \mathbf{y}_k|\mathbf{y}_{1:k-1})$?

Green: $\mathbf{P}_{k-1|k-1}, \hat{\mathbf{x}}_{k-1|k-1}$

Orange: $\mathbf{S}_k$

Pink: $\mathbf{S}_k, \mathbf{P}_{xy}, \hat{\mathbf{y}}_{k|k-1}$

Yellow: $\hat{\mathbf{y}}_{k|k-1}$

predicted density means we already have predicted state mean and state convariance

CHALMERS

Lemma

- If **x** and **y** are two Gaussian random variables with the joint probability density function

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \mathbf{P}_{xx} & \mathbf{P}_{xy} \\ \mathbf{P}_{yx} & \mathbf{P}_{yy} \end{bmatrix} \right)$$

then the conditional density of **x** given **y** is

$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\mathbf{x}; \mu_x + \mathbf{P}_{xy}\mathbf{P}_{yy}^{-1}(\mathbf{y} - \mu_y), \mathbf{P}_{xx} - \mathbf{P}_{xy}\mathbf{P}_{yy}^{-1}\mathbf{P}_{yx})$$